LABORATORY OBSERVATION / RECORD

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 3
Student Name: Gorapalli Sravya
Roll No.: A24126552257
Date: 26-08-2026

---

1. TITLE

Demonstrate JavaScript Basics Using Arrays and Functions

2. AIM

To write and execute JavaScript programs to understand the use of arrays and functions.

3. OBJECTIVE

The objectives of this experiment are:

- To understand JavaScript array declaration and array methods.
- To understand function declaration, function calling and return values.
- To perform basic array operations (sum, maximum, sorting) using functions.
- To display the computed results on a webpage.

4. THEORY

JavaScript is a scripting language used to add interactivity and logic to web pages. An array is an ordered collection of values accessed using index numbers, declared using square brackets []. Arrays support built-in methods such as push(), pop(), sort(), map() and forEach().

A function is a reusable block of code defined using the function keyword that performs a specific task. Functions can accept parameters and return a value using the return statement, which helps organize code and avoid repetition.

Functions and arrays are commonly used together in JavaScript to process collections of data, such as calculating totals, finding extreme values, or reordering elements.

5. ALGORITHM / PROCEDURE

1. Create an HTML file with an embedded or linked JavaScript file.
2. Declare an array of numbers.
3. Define a function sumArray() to calculate the sum of the array elements.
4. Define a function findMax() to find the maximum element in the array.
5. Define a function sortArray() to return a sorted copy of the array.

6. Call each function with the declared array.
7. Display the results on the webpage using the DOM.
8. Save the file and open it in a browser.
9. Check the browser console for the logged output.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>JS Arrays &amp; Functions</title>
</head>
<body>
  <h1>JavaScript Arrays &amp; Functions Demo</h1>
  <div id="output"></div>

  <script src="script.js"></script>
</body>
</html>
```

script.js

```
const numbers = [12, 45, 7, 23, 56, 3, 89];

function sumArray(arr) {
  let sum = 0;
  for (let i = 0; i < arr.length; i++) {
    sum += arr[i];
  }
  return sum;
}

function findMax(arr) {
  let max = arr[0];
  for (let i = 1; i < arr.length; i++) {
    if (arr[i] > max) {
      max = arr[i];
    }
  }
  return max;
}

function sortArray(arr) {
  return [...arr].sort((a, b) => a - b);
}

const total = sumArray(numbers);
const maximum = findMax(numbers);
const sorted = sortArray(numbers);

document.getElementById("output").innerHTML = `
  <p>Array: ${numbers.join(", ")}</p>
  <p>Sum: ${total}</p>
  <p>Maximum: ${maximum}</p>
  <p>Sorted Array: ${sorted.join(", ")}</p>
```

```
`;

console.log("Array:", numbers);
console.log("Sum:", total);
console.log("Maximum:", maximum);
console.log("Sorted:", sorted);
```

7. CODE EXPLANATION

Tag / Code	Explanation
const numbers = [...]	Declares an array of numbers
function sumArray(arr)	Calculates the sum of all elements in the array
function findMax(arr)	Finds the maximum value in the array
function sortArray(arr)	Returns a sorted copy of the array in ascending order
arr.length	Returns the number of elements in the array
for loop	Iterates through each element of the array
[...arr].sort()	Creates a copy of the array and sorts it numerically
document.getElementById()	Accesses an HTML element by its id to display output
console.log()	Prints output to the browser console

8. TEST CASES AND EXECUTION DATA

The script was run in the browser and each function's output was verified against the expected result.

Test Case	Description	Expected Result	Actual Result	Status
TC-01	Run script with sample array	Sum, max and sorted array computed correctly	All values computed correctly	Pass
TC-02	Check sumArray() function	Displays correct sum of all elements	Correct sum displayed	Pass
TC-03	Check findMax() function	Displays correct maximum value	Correct maximum displayed	Pass
TC-04	Check sortArray() function	Array displayed in ascending order	Array displayed in ascending order	Pass
TC-05	Check browser console	All values logged correctly	All values logged correctly	Pass

9. OUTPUT

Output: Result Displayed on Webpage and Console

[Insert screenshot of the webpage output and browser console showing the array, sum, maximum and sorted result.]

10. RESULT

Thus, JavaScript programs using arrays and functions were successfully written and executed, and the sum, maximum and sorted output were verified both on the webpage and in the console.